Word Codes (Set B)

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Spotting the Code

1. COG — shifting each letter BACK by 1: D→C, P→O, H→G.
2. DBU — shifting each letter forward 1: C→D, A→B, T→U.
3. A (QJH) — shifting forward 1: P→Q, I→J, G→H.
4. A, B, C, D, E — every pair listed is correctly shifted forward by 3.
5. DAT — shifting each letter BACK 1: E→D, B→A, U→T.
6. A (SVO) — shifting forward 1: R→S, U→V, N→O.
7. JGP — shifting forward 2: H→J, E→G, N→P.
8. IOG — shifting each letter BACK 2: K→I, Q→O, I→G.
9. IAT — shifting each letter BACK 1: J→I, B→A, U→T.
10. A, B, C, D — each correctly reverses the +1 shift by moving back 1; E is incorrect (K→J, Q→P, I→H gives JPH, not LPH).
11. QJH — shifting forward 1: P→Q, I→J, G→H.
12. A (AZS) — shifting back 1: B→A, A→Z (wrapping), T→S.
13. JEX — shifting each letter BACK 2: L→J, G→E, Z→X.
14. FDW — shifting forward 3: C→F, A→D, T→W.
15. A, B, C, D — each is correctly shifted forward by 2; E is incorrect (C→E, A→C, T→V, giving ECV, not EDV).

Applying the Code

16. The shift is +2 (C→E, A→C, T→V). Encoding DOG with +2: D→F, O→Q, G→I, giving the code 'FQI'.
17. CON — shifting back 4: G→C, S→O, R→N.
18. SWIM — shifting forward 3 (undoing a back-3 shift): P→S, T→W, F→I, J→M.
19. A, B, C, D, E — every pair is correctly shifted forward by 5.
20. The shift is +2 (B→D, I→K, R→T, D→F). Encoding NEST with +2: N→P, E→G, S→U, T→V, giving 'PGUV'.
21. TIN — decoding PEJ by shifting forward 4 (undoing a back-4 shift): P→T, E→I, J→N.
22. WIND — decoding ZLQG by shifting BACK 3: Z→W, L→I, Q→N, G→D.
23. BTQK — encoding WOLF with a +5 shift: W→B (wrapping), O→T, L→Q, F→K.
24. A, B, C, D, E — every pair is correctly shifted back by 6 (wrapping where needed).
25. The shift is +2 (D→F, U→W, C→E, K→M). Decoding JQWUG by shifting BACK 2: J→H, Q→O, W→U, U→S, G→E, giving 'HOUSE'.
26. LOYN — encoding FISH with a +6 shift: F→L, I→O, S→Y, H→N.
27. FISH — decoding ILVK by shifting BACK 3: I→F, L→I, V→S, K→H.
28. BIRD — decoding UBKW by shifting FORWARD 7 (undoing a back-7 shift): U→B, B→I, K→R, W→D.
29. A, B, C, D, E — every pair is correctly shifted forward by 4.
30. The shift is +3 (H→K, O→R, R→U, S→V, E→H). Encoding SNAKE with +3: S→V, N→Q, A→D, K→N, E→H, giving 'VQDNH'.

Codes in Context

31. CHAIR — the shift from TIGER to VKIGT is +2. Decoding EJCKT by −2: E→C, J→H, C→A, K→I, T→R.
32. J — G is the 7th letter; 7+3=10, and the 10th letter is J.
33. A) NGI — shifting LEG forward 2: L→N, E→G, G→I.

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34. THE — decoding WKH by shifting back 3: W→T, K→H, H→E.

35. FDW — the shift from HEN to KHQ is +3. Encoding CAT with +3: C→F, A→D, T→W.

36. A) W — reversing the alphabet: D is the 4th letter, so its partner is the (27−4)=23rd letter, which is W.

37. DOG — the shift from BAT to DCV is +2. Decoding FQI by −2: F→D, Q→O, I→G.

38. DPG — the consonants D and G stay the same; the vowel O shifts forward 1 to P.

39. A, B, C, E — each consonant shifts back 1 (C→B, T→S, confirming A and C), the vowel A is unaffected (B), and the coded word is BAS (E). D is false because A stays as A, not Z.

40. FQI — the shift from CAT to ECV is +2. Encoding DOG with +2: D→F, O→Q, G→I.

41. A) XZY — reversing the alphabet: C(3rd) ↔ X(24th), A(1st) ↔ Z(26th), B(2nd) ↔ Y(25th).

42. UBD — shifting CAT forward 1 gives DBU; reversing DBU gives UBD.

43. T — P is the 16th letter; 16+4=20, and the 20th letter is T.

44. A) WAT — the shift from DOG to GRJ is +3. Decoding ZDW by −3: Z→W, D→A, W→T.

45. HPE — reversing DOG gives GOD; shifting GOD forward 1: G→H, O→P, D→E.

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Homework Worksheet — Answers

Quick Practice

1. EAT — shifting each letter BACK by 1: F→E, B→A, U→T.
2. EPH — shifting forward 1: D→E, O→P, G→H.
3. A (SVH) — shifting forward 1: R→S, U→V, G→H.
4. A, B, C, D, E — every pair listed is correctly shifted forward by 2.
5. PIG — shifting each letter BACK 1: Q→P, J→I, H→G.
6. A (DBU) — shifting forward 1: C→D, A→B, T→U.
7. HQZ — shifting forward 2: F→H, O→Q, X→Z.
8. HEN — shifting each letter BACK 2: J→H, G→E, P→N.
9. RUN — shifting each letter BACK 1: S→R, V→U, O→N.
10. A, B, C, D — each correctly reverses the +2 shift by moving back 2; E is incorrect (E→C, D→B, V→T, giving CBT, not CAT).
11. IFO — shifting forward 1: H→I, E→F, N→O.
12. A (CNF) — shifting back 1: D→C, O→N, G→F.
13. COG — shifting each letter BACK 3: F→C, R→O, J→G.
14. RKI — shifting forward 2: P→R, I→K, G→I.
15. A, B, C, D — each pair is correctly shifted forward by 1; E is incorrect (B→C, A→B, T→U, giving CBU, not CAU).

Mixed Practice

16. The shift is +2 (D→F, O→Q, G→I). Encoding CAT with +2: C→E, A→C, T→V, giving 'ECV'.
17. WAT — shifting ZDW back 3: Z→W, D→A, W→T.
18. TIIR — shifting VKKT back 2: V→T, K→I, K→I, T→R.
19. A, B, C, D, E — every pair listed is correctly shifted forward by 4.
20. The shift is +2 (F→H, I→K, S→U, H→J). Encoding CRAB with +2: C→E, R→T, A→C, B→D, giving 'ETCD'.
21. OLUE — shifting ROXH back 3: R→O, O→L, X→U, H→E.
22. STAR — decoding VWDU by shifting BACK 3: V→S, W→T, D→A, U→R.
23. DIFVE — encoding ZEBRA with +4: Z→D (wrapping), E→I, B→F, R→V, A→E, giving 'DIFVE'.
24. A, B, C, D, E — every pair is correctly shifted back by 5 (wrapping where needed).
25. OHMER — the shift is +2 (C→E, R→U, A→C, B→D). Decoding QJOGT by shifting BACK 2: Q→O, J→H, O→M, G→E, T→R.
26. LTFY — encoding GOAT with +5: G→L, O→T, A→F, T→Y.
27. SNAKE — decoding VQDNH by shifting BACK 3: V→S, Q→N, D→A, N→K, H→E.
28. FQCRD — decoding ZKWLX by shifting FORWARD 6 (undoing a back-6 shift): Z→F, K→Q, W→C, L→R, X→D.
29. A, B, C, E — checking each with +3: BEAR→EHDU (matches A); SWAN→VZDQ (matches B); GOAT→JRDW (matches C); LION+3 gives OLRQ, not the ORLQ stated in option D, so D is wrong; DEER→GHHU (matches E).
30. The shift is +3 (M→P, O→R, U→X, S→V, E→H). Encoding TIGER with +3: T→W, I→L, G→J, E→H, R→U, giving 'WLJHU'.

Standard / Advanced

31. E — the shift from DOG to FQI is +2, so C becomes E.
32. P — the shift from CAT to ECV is +2, so N becomes P.
33. C) G — the shift from HEN to KHQ is +3, so D becomes G.
34. T — the shift from BAT to DCV is +2, so R becomes T.
35. G — the letter with position number 4 is D; 3 places after D is G.

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36. WOLF — the shift from TIGER to VKIGT is +2. Decoding YQNH by −2: Y→W, Q→O, N→L, H→F.

37. A) ULYHU — the shift from HEN to KHQ is +3. Encoding RIVER with +3: R→U, I→L, V→Y, E→H, R→U.

38. IQF — shifting DOG forward 2 gives FQI; reversing FQI gives IQF.

39. B) B — W is the 23rd letter; 23+5=28, which wraps round to 28−26=2, the letter B.

40. A, B, C, E — the consonants C and T stay the same (A, C), the vowel A shifts forward 1 to B (B), and the coded word is CBT (E). D is false because A becomes B, not Z.

41. WLT — reversing the alphabet: D(4th) ↔ W(23rd), O(15th) ↔ L(12th), G(7th) ↔ T(20th).

42. CASE — the shift from APPLE to DSSOH is +3. Decoding FDVH by −3: F→C, D→A, V→S, H→E.

43. A) UBD — reversing CAT gives TAC; shifting TAC forward 1: T→U, A→B, C→D.

44. 26 — D(4)+O(15)+G(7)=26.

45. UIHES — the consonants T, G and R each shift forward 1 (to U, H, S); the vowels I and E stay the same.

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